ASSIGNMENT 5

Deadline: 11:59 pm, April 29, 2026

Submit via Blackboard
Please follow the course policy and the school's academic honesty policy.

1 Compulsory Questions

Question 1 (30 %)

Consider the generative classifier where the class conditional densities are multivariate Gaussians:

$$p(x|y=c,\theta) = \mathcal{N}(x|\mu_c,\Sigma_c) = \frac{1}{(2\pi)^{d/2}|\Sigma_c|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_c)^T\Sigma_c^{-1}(x-\mu_c)\right),$$

where d is the dimension of each data x.

Then, the corresponding class posterior has the form:

$$p(y=c|x,\theta) \propto \pi_c\mathcal{N}(x|\mu_c,\Sigma_c),$$

where $\pi_c = p(y=c)$ is the prior probability of class c.

1. Gaussian Discriminant Analysis (GDA) model is the log posterior over class labels: $\log p(y=c|x,\theta)$. Please write the derivation of getting the GDA model's discriminant function.

   Hint: You can use "const" to represent the constant in the formula.

2. Only for this sub-question, we consider a 2-classes setting. Please provide the decision boundary equation in terms of $(\pi_c, \mu_c, \Sigma_c, c = 0, 1)$ and explain in which case the boundary will be a linear one.

3. The likelihood function of GDA is as follows:

   $$p(\mathcal{D}|\theta) = \prod_{n=1}^{N} \mathrm{Cat}(y_n|\pi) \prod_{c=1}^{C} \mathcal{N}(x_n|\mu_c,\Sigma_c)^{\mathbb{I}(y_n=c)},$$

   where $\mathcal{D} = \{(x_n, y_n) : n = 1, \ldots, N\}$ and $\mathrm{Cat}(y_n|\pi)$ is the categorical distribution (π_c denotes as the probability of $y_n = c$ and N_c denotes as the number of times the event $y_n = c$ is observed). And the indicator function $\mathbb{I}(y_n = c)$ takes the value 1 if $y_n = c$; otherwise, it equals 0.

   (a) Please derive the log-likelihood function $\log p(\mathcal{D}|\theta)$.
   Hint: A detailed derivation process is required. A result without the derivation process can gain at most 1 point. You can use the term $\mathcal{N}(x|\mu_c,\Sigma_c)$ to simplify your answer.

   (b) For $p(\mathcal{D}|\theta)$, we have $\theta = \{\pi, \mu_c, \Sigma_c\}$. Please write down the details of how to optimize and get the maximum likelihood estimator $\hat{\theta}_c$.
   Hint1: Consider minimizing the negative log likelihood (NLL) subject to the equality constraint with respect to π, using the method of Lagrange multipliers.
   Hint2: A detailed derivation process is required. A result without the derivation process can gain at most 1 point.

Question 2 (20 %)

Given the following linearly separable 3D dataset:

- **Class +1**: $(1, 0, 0)$
- **Class -1**: $(1, -\sqrt{2}, 1)$, $(1, \sqrt{2}, 1)$

Your task is to

1. Write the primal hard-margin SVM optimization problem.
2. Derive the Lagrangian and dual problem.
3. Solve the dual analytically for optimal α_i.
4. Recover the primal variables $\mathbf{w}$ and b.
5. State the decision boundary equation.

Question 3 (50 %, Coding Exercise)

In this exercise, you will implement GDA for classification. The training dataset will be provided, containing multiple data samples (x, y), where x is a feature vector and y is the corresponding class label. You are first required to implement GDA in Python, estimate the model parameters from data, and use the model for classification. After that, you are also required to implement Linear Discriminant Analysis (LDA) in python, which is a special case of GDA. After completing this exercise, you are expected to gain a better understanding of GDA and LDA, as well as the differences between them.

Data Description:

- Training data: `train.csv`
- Test data: `test.csv`
- In the dataset:
  - x is a 2-dimensional feature vector, corresponding to the columns $x1$ and $x2$,
  - y is the class label, taking values in $\{0, 1\}$, corresponding to the column y. Only the training data contains values for y. The test data does not contain the column y and you need to predict the labels for the test data.

Task Description:

1. Use the training data to fit a GDA model via maximum likelihood estimation. You need to estimate the class prior probabilities, the class means, and the class covariance matrices.
2. Use the fitted GDA model to classify the test data and save the classification results.
3. Compute the decision boundary and plot the classification results of the GDA model on the test data. The plot should include all data points and the decision boundary. Different classes of data points should be distinguished using different representations (e.g., different colors).
4. Use the training data to fit an LDA model via MLE. Note that LDA is a special case of GDA in which all classes share the same covariance matrix. Therefore, you can use all data to estimate a shared covariance matrix.
5. Use the fitted LDA model to classify the test data and save the classification results.
6. Compute the decision boundary and plot the classification results of the LDA model on the test data. The requirements for the plot are the same as those for the GDA model.

Submission Requirements:

- **A Python file (e.g., `gda.py`) containing your implementation.**
- **Two prediction result files (one for each model). You should fill in the predicted labels for the test data and save them in two separate files (e.g., `gda_predictions.csv` and `lda_predictions.csv`).**
- **Two classification plots (one for each model).**

Additional Notes:

You may use libraries like `numpy`, `pandas` and `matplotlib` for numerical computations and plotting. Any built-in classification models are `forbidden` in this exercise (e.g. `LinearDiscriminantAnalysis` in `sklearn(Scikit-learn)`).

2 Optional Questions

Question 4

In a certain town, 60% of taxis are green and 40% are white. Because of poor weather conditions of this town, a witness correctly identifies the color of a taxi with probability 80%. One day, the witness reports seeing a white taxi. What is the probability that the taxi was actually white?

Question 5

Consider a sequence of independent Bernoulli trials, each with the same unknown probability of success probability θ. Let $x_1, x_2, \cdots, x_n$ be the observed outcomes, where $x_i = 1$ if the i-th trial is a success, and $x_i = 0$ if it is a failure.

1. Write the maximum likelihood estimate (MLE) of θ based on the observed data $x_1, x_2, \cdots, x_n$.
2. Use the MLE to estimate the success rate θ based on the observed data.

Hint: The probability density function of Bernoulli distribution is $f(x) = \theta^x(1-\theta)^{(1-x)}$, where θ is the success rate.

Question 6

The discriminant function for Gaussian Discriminant Analysis (GDA) is given as:

$$\log p(y = c|\mathbf{x}, \theta) = \log \pi_c - \frac{1}{2}\log|2\pi\Sigma_c| - \frac{1}{2}(\mathbf{x} - \mu_c)^T\Sigma_c^{-1}(\mathbf{x} - \mu_c) + \text{const},$$

where π_c is the prior probability of class c, $\mathbf{x}$ denotes the data feature, θ represents the model parameters, μ_c and Σ are the mean and covariance matrix of class c.

1. Please write the decision boundary between two classes c and c'.
2. LDA (Linear Discriminant Analysis) is a special case of GDA where the class covariances are assumed to be equal, i.e.,
   $$\Sigma_c = \Sigma, \quad \forall c.$$
   Please write the decision boundary between two classes c and c'.

Question 7

We have a dataset for 2 classes.

- For **class 1**, we have these samples:
  $$\{(2.1, 1.2, 0.3), (0.9, -0.8, -0.1), (0.2, -1.5, -1.8), (1.5, -0.4, -0.7)\}.$$
- For **class 2**, we have these samples:
  $$\{(-0.7, -0.6, -0.9), (-1.0, -1.2, -0.3), (-1.5, 0.4, -1.1), (-0.6, -0.5, 1.0)\}.$$

We want to fit a **Fisher's LDA model** to this dataset. For this question, all results should be rounded to 3 decimal places.

1. Calculate the mean of the two classes, $\mu_\mathbf{1}$ and $\mu_\mathbf{2}$.
2. Calculate the between-class scatter matrix $\mathbf{S_B}$.
3. Calculate the within-class scatter matrix $\mathbf{S_W}$.
4. Calculate the optimal projection vector $\mathbf{w}^\star$ based on $\mathbf{S_B}$ and $\mathbf{S_W}$.

Homework Submission Policy

- **Full Points:** At most 8 points in total if submitted before the deadline.
- **Late Submission:** At most 6.4 points in total if submitted within 24 hours after the deadline.

- **No Points:** 0 points if submitted more than 24 hours after the deadline.
- You are encouraged to either type your solution in LaTeX or Word, You can also scan or take photos of your handwritten solution and bundle your scans/photos into one single PDF file.
- All written answers should be submitted in **a single PDF file**. Please also compress the **related codes and plots for question 3 into a Zip file** and submit it along with your PDF answer.
- Please submit via Blackboard.

Question 1

1) $p(y=c|x,\theta) \propto \pi_c N(x|\mu_c, \Sigma_c)$

$\ln p(y=c|x,\theta) = \ln \pi_c + \ln N(x|\mu_c \Sigma_c) + const$

$\because N(x|\mu_c, \Sigma_c) = \frac{1}{2\pi^{\frac{d}{2}} |\Sigma_c|^{\frac{1}{2}}} \exp(-\frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c))$

$\ln N(x|\mu_c, \Sigma_c) = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_c| - \frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c)$

$\therefore \ln p(y=c|x,\theta) = \ln \pi_c - \frac{1}{2}\ln|\Sigma_c| - \frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c) + const$

Let $f_c(x) = \ln p(y=c|x,\theta)$

$f_c(x) = \ln \pi_c - \frac{1}{2}\ln|\Sigma_c| - \frac{1}{2}(x-\mu_c)^T \Sigma_c^{-1}(x-\mu_c) + const.$

2) $f_0(x) = f_1(x)$

$\ln \pi_1 - \frac{1}{2}\ln|\Sigma_1| - \frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1) = \ln \pi_0 - \frac{1}{2}\ln|\Sigma_0| - \frac{1}{2}(x-\mu_0)^T \Sigma_0^{-1}(x-\mu_0)$

$\ln \frac{\pi_1}{\pi_0} - \frac{1}{2}\ln\frac{|\Sigma_1|}{|\Sigma_0|} - \frac{1}{2}[(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1) - (x-\mu_0)^T \Sigma_0^{-1}(x-\mu_0)] = 0$

In common ($\Sigma_0 \neq \Sigma_1$), the eq. will have x^TAx term, it will be a quadratic decision boundary

But if $\Sigma_0 = \Sigma_1$, the quadratic form will be cancelled each others, it becomes $w^Tx + b = 0$ and the eq. will be a linear decision boundary

i.e.

$\ln\frac{\pi_1}{\pi_0} - \frac{1}{2}\ln 1 - \frac{1}{2}[(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1) - (x-\mu_0)^T \Sigma_0^{-1}(x-\mu_0)] = 0$

$\ln\frac{\pi_1}{\pi_0} - \frac{1}{2}[(x^T\Sigma_1^{-1}x - 2\mu_1^T\Sigma_1^{-1}x + \mu_1^T\Sigma_1^{-1}\mu_1) - (x^T\Sigma_0^{-1}x - 2\mu_0^T\Sigma_0^{-1}x + \mu_0^T\Sigma_0^{-1}\mu_0)] = 0$

$\ln\frac{\pi_1}{\pi_0} - \frac{1}{2}[2(\mu_0-\mu_1)^T\Sigma^{-1}x + (\mu_1^T\Sigma^{-1}\mu_1 - \mu_0^T\Sigma^{-1}\mu_0)] = 0$

Question 1

3a) $p(D|\theta) = \prod_{n=1}^{N} Cat(y_n|\pi) \prod_{c=1}^{C} N(x_n|\mu_c, \Sigma_c)^{\mathbb{I}(y_n=c)}$

$\ln p(D|\theta) = \ln\left(\prod_{n=1}^{N} Cat(y_n|\pi)\right) + \ln\left(\prod_{n=1}^{N}\prod_{c=1}^{C} N(x_n|\mu_c, \Sigma_c)^{\mathbb{I}(y_n=c)}\right)$

$= \sum_{n=1}^{N} \ln Cat(y_n|\pi) + \sum_{n=1}^{N}\sum_{c=1}^{C} \mathbb{I}(y_n=c) \ln N(x_n|\mu_c, \Sigma_c)$

$= \sum_{n=1}^{N}\sum_{c=1}^{C} \mathbb{I}(y_n=c) \ln \pi_c + \sum_{n=1}^{N}\sum_{c=1}^{C} \mathbb{I}(y_n=c) \ln N(x_n|\mu_c, \Sigma_c)$

3b) $\hat{\theta}_c = \{\hat{\pi}_c, \hat{\mu}_c, \hat{\Sigma}_c\}$

$\sum_{c=1}^{C} \pi_c = 1$

$L(\pi, \lambda) = \sum_{c=1}^{C} N_c \ln \pi_c + \lambda(1 - \sum_{c=1}^{C} \pi_c)$

$\frac{\partial L}{\partial \pi_c} = \frac{N_c}{\pi_c} - \lambda = 0$

$\pi_c = \frac{N_c}{\lambda}$

Sub $\sum_c \pi_c = 1$,

$\sum_c \frac{N_c}{\lambda} = 1$

$\lambda = \sum_c N_c$

$\lambda = N$

$\therefore \hat{\pi}_c = \frac{N_c}{N}$

$\frac{\partial}{\partial \mu_c} \sum_{n:y_n=c} \left(-\frac{1}{2}(x_n - \mu_c)^T \Sigma_c^{-1} (x_n - \mu_c)\right) = 0$

$\sum_{n:y_n=c} \Sigma_c^{-1} (x_n - \mu_c) = 0$

$\therefore \hat{\mu}_c = \frac{1}{N_c} \sum_{n:y_n=c} x_n$

$\hat{\Sigma}_c = \frac{1}{N_c} \sum_{n:y_n=c} (x_n - \hat{\mu}_c)(x_n - \hat{\mu}_c)^T$

$\therefore \hat{\theta}_c \left\{ \frac{N_c}{N}, \frac{1}{N_c} \sum_{n:y_n=c} x_n, \frac{1}{N_c} \sum_{n:y_n=c} (x_n - \mu_c)(x_n - \mu_c)^T \right)$

Question 2

Dataset :

Class $+1$: $x_1 (1, 0, 0)$ $(y_1 = +1)$

Class -1 : $x_2 (1, -\sqrt{2}, 1)$ $(y_2 = -1)$, $x_3 (1, \sqrt{2}, 1)$ $(y_3 = -1)$

1) Primal Hard - margin SVM Optimization Problem

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

$$y_n (w^T x_i + b) \geq 1 \text{ , } n = 1, 2, 3$$

i.e.

$$\begin{cases} w^T x_1 + b \geq 1 \\ -(w^T x_2 + b) \geq 1 \\ -(w^T x_3 + b) \geq 1 \end{cases}$$

2) Lagrangian and Dual Problem

$$L(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^{3} \alpha_i [y_i (w^T x_n + b) - 1] \quad \forall\ \alpha_i \geq 0$$

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{3} \alpha_i y_i x_i = 0$$

$$\therefore\ w = \sum_{i=1}^{3} \alpha_i y_i x_i$$

$$\sum_{i=1}^{3} \alpha_i y_i = 0$$

$$\alpha_1 - \alpha_2 - \alpha_3 = 0$$

$$\alpha_1 = \alpha_2 + \alpha_3$$

$$\therefore\ \max_{\alpha} \sum_{i=1}^{3} \alpha_i - \frac{1}{2} \sum_{i=1}^{3} \sum_{j=1}^{3} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

$$\text{s.t. } \alpha_1 = \alpha_2 + \alpha_3 \text{ , } \alpha_i \geq 0$$

Question 2

3) Dual analytically

$x_1^T x_1 = 1$

$x_2^T x_2 = 1 + 2 + 1 = 4$

$x_3^T x_3 = 1 + 2 + 1 = 4$

$x_1^T x_2 = 1$

$x_1^T x_3 = 1$

$x_2^T x_3 = 1 - 2 + 1 = 0$

$$L = \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2}[\alpha_1^2 (+1)^2 (1) + \alpha_2^2 (-1)^2 (4) + \alpha_3^2 (-1)^2 (4)$$
$$+ 2\alpha_1\alpha_2 (+1)(-1)(1) + 2\alpha_1\alpha_3 (+1)(-1)(1) + 2\alpha_2\alpha_3 (-1)(-1)(0)]$$

$$= \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2}[\alpha_1^2 + 4\alpha_2^2 + 4\alpha_3^2 - 2\alpha_1\alpha_2 - 2\alpha_1\alpha_3]$$

$\because$ $\alpha_1 = \alpha_2 + \alpha_3$ and x_2, x_3 are symmetric.

$\therefore$ $\alpha_2 = \alpha_3$.

Let $\alpha_1 = 2\alpha$, $\alpha_2 = \alpha_3 = \alpha$

$\therefore L = 4\alpha - \frac{1}{2}(12\alpha^2 - 8\alpha^2) = 4\alpha - 2\alpha^2$

$\frac{\partial L}{\partial \alpha} = 4 - 4\alpha = 0$

$\alpha = 1$

$\therefore$ $\alpha_1 = 2$, $\alpha_2 = \alpha_3 = 1$.

Question 2

4) Primal var. w, b.

$w = \sum_{i=1}^{3} \alpha_i y_i x_i$

$= 2(1,0,0) - (1,-\sqrt{2},1) - (1,\sqrt{2},1)$

$= (0,0,-2)$

Use support vector x_1

$y_1(w^T x_1 + b) = 1$

$(0,0,-2)\cdot(1,0,0) + b = 1$

$b = 1$

5) Decision boundary eq.

$w^T x + b = 0$

$-2x_3 + 1 = 0$

$x_3 = \frac{1}{2}$

Attachment

Declaration for written assignment

I am/we are submitting the assignment for:
☑ an individual project or
□ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged/all members of the group have read and checked that all parts of the piece of work, irrespective of whether they are contributed by individual members or all members as a group, here submitted are original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I/We also acknowledge that I am/we are aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

In the case of a group project, we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic content.

I/we declare that I/we have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I/we declare that I/we have read and understood the University's policy on the use of AI for academic work. I/we confirm that I/we have complied with the instructions given by my/our course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my/our submission.

I/We also understand that assignments without a properly signed declaration by the student concerned and in the case of a group project, by all members of the group concerned, will not be graded by the teacher(s).

Signature(s)

29/4/2025
Date

Fung Cheuk Nam
Name~~(s)~~

1155192095
Student ID(s)

CSCI 3320 -
Course code

Fundamental of Machine Learning
Course title

Use of AI tools.

I acknowledge the use of ChatGPT, Gemini, Claude (2026, Apr 29)

for content guidance, code guidance